

TASK 2 — MOTOR CONTROL WITH PWM & OBSTACLE DETECTION

Arduino UNO + L293D + DC Motor + HC-SR04

1. Goal

Control a DC motor's speed using PWM and automatically stop the motor when an HC-SR04 ultrasonic sensor detects an obstacle within a specified distance.

2. Components Required

- Arduino UNO
- L293D motor driver
- DC motor
- HC-SR04 ultrasonic sensor
- External motor power supply
- Breadboard and jumper wires

3. Wiring Connections

L293D / Function	Arduino UNO
EN1 (Enable 1)	D5 (PWM)
IN1	D7
IN2	D8
Logic VCC	5V
GND	GND
Motor OUT1 / OUT2	DC Motor
Motor VCC	External motor supply

HC-SR04	Arduino UNO
VCC	5V
GND	GND
TRIG	D9
ECHO	D10

Important: Connect Arduino GND and motor-driver GND together. Use a suitable external supply for the motor; do not drive the motor directly from an Arduino I/O pin.

4. Arduino Program

```
// Task 2: Motor Control with PWM & Obstacle Detection

#define ENA 5
#define IN1 7
#define IN2 8

#define TRIG 9
#define ECHO 10

int motorSpeed = 180;
int obstacleDistance = 20;

void setup()
{
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  Serial.begin(9600);

  stopMotor();
}

void loop()
{
  long duration;
  float distance;

  digitalWrite(TRIG, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG, HIGH);
  delayMicroseconds(10);

  digitalWrite(TRIG, LOW);

  duration = pulseIn(ECHO, HIGH);

  distance = duration * 0.0343 / 2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  if (distance <= obstacleDistance && distance > 0)
  {
    stopMotor();
    Serial.println("Obstacle detected - Motor STOPPED");
  }
  else
  {
    runMotor(motorSpeed);
    Serial.println("Path clear - Motor RUNNING");
  }

  delay(200);
}

void runMotor(int speedValue)
{
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, speedValue);
}

void stopMotor()
{
  analogWrite(ENA, 0);
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
}
```

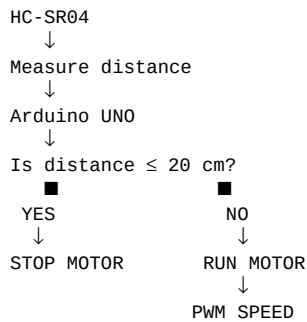
5. Program Explanation

PWM: D5 is a hardware PWM pin on Arduino UNO. The value passed to analogWrite() controls the motor driver's enable signal. A value of 0 stops the motor and 255 gives the maximum PWM duty cycle.

Ultrasonic sensing: The HC-SR04 sends an ultrasonic pulse through TRIG and receives the reflection through ECHO. Arduino measures the echo time and calculates distance using: **Distance = time × speed of sound ÷ 2**.

Obstacle decision: If the measured distance is 20 cm or less, the motor is stopped. If the path is clear, the motor runs at the selected PWM speed.

6. Working Flow



7. Expected Serial Monitor Output

Distance: 45.2 cm
Path clear - Motor RUNNING

Distance: 18.5 cm
Obstacle detected - Motor STOPPED

8. Test Results — Distance vs Response

Distance	Motor Response
50 cm	RUN
40 cm	RUN
30 cm	RUN
25 cm	RUN
20 cm	STOP
15 cm	STOP
10 cm	STOP
5 cm	STOP

The stopping distance can be changed in the program by modifying **obstacleDistance**. For example, setting it to 15 makes the motor stop at approximately 15 cm or closer.

9. Result

The system successfully controls DC motor speed using PWM and automatically stops the motor when the HC-SR04 ultrasonic sensor detects an obstacle within the specified distance.

10. Deliverables Checklist

- Arduino code implementing PWM speed control and obstacle stop
- Wiring diagram/connections

- Brief test results showing distance versus motor response
- Demonstration video, if required